CloudGuard Pro

CSPM v3

User & Configuration Guide

Multi-Cloud Security Posture Management

Aniza Corp • Version 3.0.0 • 2026
Supports: AWS • Azure • GCP • IBM Cloud • Oracle Cloud

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1. Introduction

CloudGuard Pro is a Cloud Security Posture Management (CSPM) platform built by Aniza Corp. It continuously evaluates your cloud infrastructure against industry security benchmarks and highlights misconfigured resources before they become security incidents.

1.1 What CloudGuard Pro Does

CloudGuard Pro connects to your cloud provider accounts and performs read-only checks against your resources. It produces a list of security findings, each mapped to a recognised security standard, with clear remediation steps.

- Scans AWS, Microsoft Azure, Google Cloud Platform, IBM Cloud, and Oracle Cloud Infrastructure
- Runs 52 built-in security checks mapped to CIS Benchmarks, NIST, OWASP, PCI DSS, ISO 27001, and more
- Produces findings with severity ratings (Critical, High, Medium, Low, Informational)
- Lets you create, import, and export your own custom security checks
- Generates PDF security reports for management and audit purposes

1.2 Who This Guide Is For

This guide is written for IT administrators and security analysts who are responsible for cloud security. No programming knowledge is required to use CloudGuard Pro. All operations are performed through the web browser interface.

1.3 Key Concepts

Term	What It Means
Connection	Your cloud account credentials — tells CloudGuard Pro which account to scan
Scan	A security assessment run against a cloud account
Finding	A specific security issue detected during a scan, with severity and remediation steps
Check	A single security rule (e.g. 'Root MFA must be enabled')

Term	What It Means
Family	A group of related checks (e.g. 'Identity & Access', 'Networking')
Demo Mode	A built-in mode that runs checks against sample data — no cloud credentials needed
Suppression	Marking a finding as accepted or excluded from reporting

2. Installation

CloudGuard Pro runs on your own machine or server. There is no cloud service to sign up for — your credentials never leave your environment.

2.1 System Requirements

Component	Requirement
Operating System	macOS 12+, Ubuntu 20.04+, or any Linux distribution
Python	Version 3.9 or higher (3.13 recommended). Download from python.org
RAM	4 GB minimum, 8 GB recommended
Disk Space	500 MB for the application, plus space for scan data
Network	Outbound HTTPS (port 443) to your cloud provider APIs
Browser	Any modern browser: Chrome, Firefox, Safari, Edge

> ℹ️ **Note: Python Version**
>
> If you are on Python 3.14, the deploy script automatically sets the required compatibility flag.
>
> You do not need to do anything extra — just run the deploy command as shown below.

2.2 Extracting the Package

You will receive CloudGuard Pro as a .tar.gz archive file. Follow these steps to extract it:

1. Open a Terminal (macOS: Applications → Utilities → Terminal)
2. Navigate to the folder where you saved the archive:

```
cd ~/Downloads
```

3. Extract the package:

```
tar -xzf cloudguard-pro-cspm-v3.0.0.tar.gz
```

4. Move into the extracted folder:

```
cd cspm
```

2.3 Running the Deploy Script

The deploy script installs all required Python dependencies. Run it once before starting CloudGuard Pro for the first time, and again whenever you install a new version.

```
# Standard installation
./scripts/deploy.sh native


# Install without demo data
./scripts/deploy.sh native --no-demo
```

ℹ️ **Demo Data**

By default, CloudGuard Pro pre-loads 40 sample cloud assets and ~50 findings so you can explore the

interface immediately. These are clearly labelled as demo data.

Use --no-demo if you want to start with a clean, empty database.

You can also toggle this in the .env file: set SEED_DEMO_DATA=false

2.4 Clearing Old Build Files (Existing Installations)

If you are updating from a previous version, run the update script first to remove old build artifacts that could cause the browser to show a blank or outdated interface:

```
./scripts/update.sh
```

3. Starting and Stopping CloudGuard Pro

3.1 Starting the Application

CloudGuard Pro runs as a background server process. Start it with:

```
./scripts/start.sh native
```

Once started, you will see a confirmation like:

```
[OK]    CloudGuard Pro started (PID 12345)


  Dashboard:  http://localhost:8000

  API docs:   http://localhost:8000/docs

  Log file:   logs/cloudguard.log

  Stop:       ./scripts/stop.sh native
```

Open your web browser and go to: http://localhost:8000

3.2 Running in the Foreground

If you want to see log output directly in your terminal (useful for troubleshooting), add the --foreground flag:

```
./scripts/start.sh native --foreground
```

Press Ctrl+C to stop when running in foreground mode.

3.3 Stopping the Application

```
./scripts/stop.sh native
```

3.4 Log Files

CloudGuard Pro writes two log files in the logs/ folder:

Log File	What It Contains
logs/cloudguard.log	Main application log — startup, API requests, errors
logs/scans.log	Detailed scan log — every resource collected, every finding created, any failures

Both files rotate automatically when they reach 100 MB. Up to 5 old log files are kept.

To follow the scan log in real time:

```
tail -f logs/scans.log
```

4. Configuration

4.1 The .env File

All configuration is stored in a file called .env in the cspm folder. This file is created automatically from the template when you first run the deploy script. Open it in any text editor to make changes.

Setting	Default	Description
PORT	8000	The port the web interface runs on
HOST	0.0.0.0	The network address to listen on
SEED_DEMO_DATA	true	Set to false to disable demo data on first startup
LOG_LEVEL	info	Verbosity: debug, info, warning, error
LOG_DIR	logs	Folder where log files are stored
DATABASE_URL	(SQLite)	Database connection. Leave blank for the built-in SQLite database

⚠️ **After Editing .env**

You must stop and restart CloudGuard Pro for changes to take effect:

```
./scripts/stop.sh native
./scripts/start.sh native
```

4.2 Disabling Demo Data

The demo data (40 sample assets and ~50 findings) is useful for exploring the interface. When you are ready to use CloudGuard Pro with real cloud accounts, you may want to remove it.

Option A: Disable before first startup

Edit .env and set:

```
SEED_DEMO_DATA=false
```

Option B: Suppress demo findings in the interface

On the Findings or Assets page, use the Hide demo data toggle to hide demo items from view. You can also click Suppress all demo findings to permanently mark all demo findings as suppressed.

Option C: Start fresh

Stop the application, delete the database file, and restart:

```
./scripts/stop.sh native

rm cspm.db

./scripts/start.sh native
```

5. Connecting Your Cloud Accounts

Before running a scan, you must add a Connection for each cloud account you want to assess. A Connection tells CloudGuard Pro which account to access and how to authenticate.

Go to Connections in the left navigation menu, then click + Add Connection.

5.1 Credential Types

CloudGuard Pro supports several ways to provide credentials:

Type	How It Works	Best For
env	Reads credentials from environment variables	CI/CD pipelines, servers with IAM roles
file	Reads a credentials file you specify	Local workstations with existing credential files
profile	Uses a named profile from your credentials file	AWS users with multiple accounts
role	Assumes an IAM role (AWS) or uses a role ARN	Cross-account scanning, least-privilege setups
workload_identity	Uses the platform's built-in identity	Cloud VMs with managed identity enabled

5.2 AWS Connection Setup

Using your ~/.aws/credentials file

If you have the AWS CLI configured, your credentials are already in ~/.aws/credentials. Use them directly:

- **Credential Type:** `file`

- **Credential Ref:** `~/.aws/credentials`
- **Regions:** e.g. us-east-1, us-west-2 (blank = us-east-1 only)

> ℹ️ **About Regions**
>
> CloudGuard Pro scans only the regions you specify. If you leave Regions blank, it defaults to us-east-1.
>
> Add all regions where you have resources, separated by commas: us-east-1, eu-west-1, ap-southeast-2
>
> Scanning more regions takes longer. Start with your primary region.

Using environment variables

Set these in your .env file or export them in your terminal before starting:

```
AWS_ACCESS_KEY_ID=AKIAIOSFODNN7EXAMPLE
AWS_SECRET_ACCESS_KEY=wJalrXUtnFEMI/K7MDENG/bPxRfiCYEXAMPLEKEY
AWS_DEFAULT_REGION=us-east-1
```

> ⚠️ **Required AWS Permissions**
>
> The AWS user or role must have read-only access to the services CloudGuard Pro scans.
>
> Attach the AWS-managed policy ReadOnlyAccess, or create a custom policy with:
>
> iam:Get*, iam:List* | s3:Get*, s3:List* | ec2:Describe* | rds:Describe*
>
> cloudtrail:Describe*, cloudtrail:Get* | kms:List*, kms:Describe* | eks:List*, eks:Describe*
>
> ecr:Describe*, ecr:List* | secretsmanager:List* | lambda:List*, lambda:Get*

5.3 Azure Connection Setup

Azure uses Service Principal credentials. Create one in the Azure Portal under Azure Active Directory → App Registrations, then assign the Reader role to it.

Using a JSON credentials file

Create a text file (e.g. ~/.cloudguard/azure-creds.json) with this content:

```
{
  "tenant_id":      "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "client_id":      "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "client_secret":  "your-client-secret-value",
  "subscription_id": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx"
}
```

- **Credential Type:** `file`
- **Credential Ref:** `~/.cloudguard/azure-creds.json`

Using environment variables

```
AZURE_TENANT_ID=xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx
AZURE_CLIENT_ID=xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx
AZURE_CLIENT_SECRET=your-client-secret-value
AZURE_SUBSCRIPTION_ID=xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx
```

5.4 Google Cloud Platform (GCP) Setup

GCP uses Service Account key files. Generate one from Google Cloud Console → IAM & Admin → Service Accounts → [your account] → Keys → Add Key → JSON.

- **Credential Type:** `file`
- **Credential Ref:** `/path/to/service-account-key.json`

The service account needs the Viewer or Security Reviewer role at the project level.

5.5 Editing an Existing Connection

To update credentials, regions, or the name of an existing connection:

5. Go to Connections in the navigation menu
6. Find the connection you want to update
7. Click the ✎ Edit button on that row
8. Update the fields in the modal that appears
9. Click Save Changes

Changes take effect on the next scan. No restart is required.

6. Running Security Scans

6.1 Launching a Scan

10. Click Scans in the left navigation menu
11. In the Launch Scan section, check the box next to the connection(s) you want to scan
12. Click the ▶ Scan button

The scan status changes to running. A pulsing green dot appears in the scan history header while the scan is active. The Findings page will automatically show new results within 5 seconds of the scan completing.

6.2 Demo Scans vs Live Scans

Mode	When It Runs	What It Scans
Demo	When no cloud credentials are configured for the connection	The 40 built-in sample assets — no cloud account needed
Live	When valid credentials are detected	Your actual cloud resources via the cloud provider APIs

ℹ️ **Demo Scan**

The demo scan is a great way to explore findings and reports without needing a cloud account.

It runs immediately and produces ~50 real findings using built-in sample data.

Look for the 'demo' badge in the scan history to identify demo scans.

6.3 How Long Scans Take

Scan duration depends on how many regions you are scanning and how many resources exist in your account:

Scope	Approximate Time
Demo scan (no credentials)	5–15 seconds

Scope	Approximate Time
Single AWS region, small account	30–90 seconds
Single AWS region, large account	2–5 minutes
Multiple regions	Add ~30–60 seconds per additional region

> ℹ️ **Tip: Specify Regions**
>
> By default, CloudGuard Pro scans us-east-1 only.
>
> Add your specific regions in the connection settings to include all your resources.
>
> Scanning a single region is much faster than scanning all regions.

Scans have a 30-minute hard timeout. If a scan exceeds this, it will be marked as failed. This can happen if the cloud API is unresponsive. Check the scan log for details.

6.4 Monitoring a Running Scan

Watch the scan log file in real time to see exactly which resources are being collected:

```
tail -f logs/scans.log
```

You will see output like:

```
[scan:c5c79b4e] Live scan: collecting from AWS (aws)...

[scan:c5c79b4e] AWS: collecting IAM (global)...

[scan:c5c79b4e] AWS: IAM — 3 users

[scan:c5c79b4e] AWS: [us-east-1] S3...

[scan:c5c79b4e] AWS: S3 — 5 buckets

[scan:c5c79b4e]   NEW finding: aws-s3-001 on my-prod-bucket

[scan:c5c79b4e] Live scan complete: assets=48 checks=480 new_findings=7 resolved=0
```

7. Working with Findings

7.1 The Findings Page

The Findings page lists all security issues detected across your cloud accounts. Each finding shows:

Column	Description
Severity	Critical, High, Medium, Low, or Informational
Title	The human-readable name of the security issue
Provider	Which cloud provider the finding is from (AWS, Azure, GCP, IBM, OCI)
Service	The cloud service (e.g. iam, s3, ec2)
Resource	The specific resource that failed the check
Status	Open, Resolved, Suppressed, or Risk Accepted
First Seen	When the issue was first detected

7.2 Filtering and Searching

Use the toolbar at the top of the Findings page to narrow down results:

- Search box: find findings by title, check ID, or resource name
- All Severities: filter to Critical, High, Medium, Low, or Informational
- All Providers: show findings from one cloud provider only
- All Statuses: filter by Open, Resolved, Suppressed, or Risk Accepted
- All Families: filter by check family (e.g. Identity & Access, Networking)

7.3 Sorting Columns

Click any column header to sort by that column. Click again to reverse the sort order. The sorted column shows an arrow (▲ or ▼) to indicate direction.

7.4 Viewing Finding Details

Click any row in the findings table to open the full detail panel, which shows:

- The affected resource and all its identifiers (ARN, URN, region, account)
- A description of the security issue and why it matters
- Exact remediation steps to fix the problem
- The compliance frameworks this issue maps to (CIS, NIST, PCI DSS, etc.)
- Evidence — the specific configuration values that triggered the finding

7.5 Suppressing a Finding

If a finding is a known exception (e.g. a test account, or a compensating control exists), you can suppress it:

13. Click the finding row to open details, then click Suppress
14. Enter a reason explaining why this finding is being suppressed
15. Optionally tick Mark as Risk Accepted if the risk has been formally accepted
16. Click Confirm — the finding status changes to Suppressed

> ℹ️ **Suppressed Findings**
>
> Suppressed findings remain visible when you select 'All Statuses' in the filter.
>
> They are excluded from the Open count on the Dashboard.
>
> Use 'Risk Accepted' to distinguish findings that were formally reviewed.

7.6 Hiding Demo Data

If you have demo data loaded, use the Hide demo data toggle at the top of the Findings and Assets pages to hide all demo items from view. You can also click Suppress all demo findings to permanently mark all demo findings as suppressed.

7.7 Column Picker

Click the Columns button in the toolbar to choose which columns appear in the table. You can add or remove columns to suit your workflow. Your selection is remembered during the session.

8. Asset Inventory

The Assets page shows every cloud resource that CloudGuard Pro has discovered. Click any row to see the resource's native identifiers, cloud-specific IDs, and any open findings associated with it.

8.1 Sorting and Filtering

Use the search box to find assets by name or ID. Use the All Providers dropdown to filter by cloud provider. Click any column header to sort.

8.2 Asset Detail

Clicking a row opens the asset detail panel showing:

- Provider, service, resource type, and region
- Cloud-native identifiers: ARN (AWS), Azure Resource ID, GCP Resource Name, OCI OCID, IBM CRN
- Universal Resource Name (URN) — CloudGuard Pro's cross-cloud unique identifier
- All open findings for that specific resource

9. Check Catalog

The Check Catalog lists all 52 built-in security checks plus any custom checks you have created. Navigate to Check Catalog in the left menu.

9.1 Built-in Tab

Browse and search all built-in checks. Use the family sidebar on the left to filter by check family. Click the </> Code button on any check to view its implementation, YAML definition, provenance, and test cases.

9.2 Custom Checks Tab

Create, edit, and delete your own security checks.

Creating a Custom Check

17. Click the ✏ Custom tab then click + New Custom Check
18. Fill in the required fields: Check ID, Name, Provider, Service, Resource Type, Family, Severity
19. Add a Description explaining what the check evaluates
20. Add Remediation steps explaining how to fix a failure
21. (Optional) Add Python Implementation Code — a function that returns (passed: bool, evidence: dict)
22. (Optional) Add Compliance Mappings to link to standards like CIS or NIST
23. Click Create Check

Custom Check Logic

If you provide an Implementation Code function, it will be executed during scans.

The function must be named check(resource) and return a tuple: (True/False, {evidence dict})

Example:

```
def check(r):
    encrypted = r.get('StorageEncrypted', False)
    return encrypted, {'StorageEncrypted': encrypted}
```

Custom checks without code are saved but always return a passing result.

9.3 Importing and Exporting Checks

You can share check libraries between installations using the import and export functions.

Exporting all checks

Click ↓ Export JSON or ↓ Export YAML to download all checks (built-in and custom) as a file.

Importing checks

24. Click ↑ Import Checks in the Check Catalog toolbar
25. Upload a .json or .yaml file, or paste the content directly into the text area
26. Select the format (JSON or YAML)
27. Tick Overwrite existing if you want to replace checks with the same ID
28. Click Import Checks

Imported checks are saved to the custom checks store (backend/custom_checks/store/).

9.4 Family Files

Built-in checks are organised into family files stored in backend/checkpacks/families/{provider}/{family}.py. You can replace an individual family file to update a specific group of checks without affecting others.

Updating a family of checks

29. Drop the new .py file into backend/checkpacks/families/{provider}/
30. Click the Family Files tab in the Check Catalog
31. Click ⟳ Check for Updates

CloudGuard Pro detects and reloads changed files without restarting. The Family Files tab shows whether each file is loaded, stale, or not yet loaded.

9.5 Check Families and Frameworks

Provider	Source Frameworks	Check Count
AWS	CIS AWS Foundations v1.5.0, AWS Foundational Security Best Practices	20
Azure	CIS Azure v2.0.0, Microsoft Defender for Cloud recommendations	13
GCP	CIS GCP v2.0.0, Google Security Command Center	8
IBM Cloud	IBM Cloud Security Best Practices	5
Oracle Cloud	CIS OCI v2.0.0, OCI Cloud Guard	6

10. Reports

CloudGuard Pro can generate PDF reports suitable for management briefings, compliance audits, and security reviews. Go to Reports in the navigation menu.

10.1 Report Types

Report	Contents	Best For
Executive Summary	KPIs, severity breakdown, provider distribution, top findings	Management briefings
Technical Findings	All findings with full resource IDs, evidence, and remediation steps	Security team review
Compliance Report	Findings grouped by compliance framework (CIS, NIST, SOC 2)	Audit preparation
Asset Inventory	All assets with native identifiers and URNs	Asset management
Check Catalog PDF	All 52 checks with provenance and framework mappings	Documentation

10.2 Generating a Report

32. (Optional) Use the Report Filters to limit the report to one provider or severity level
33. Click + Generate PDF next to the report type you want
34. Wait a few seconds for the report to generate
35. Click ↓ PDF in the Report History table to download it

> ℹ️ **Report Filters**
>
> Setting a Provider filter generates a report containing only findings from that cloud provider.
>
> Setting a Severity filter excludes findings below that severity level.
>
> Leave both blank to include all providers and severities.

11. Dashboard

The Dashboard provides a high-level summary of your security posture at a glance. It refreshes automatically every 30 seconds.

Panel	Description
KPI Cards	Total open findings broken down by severity, plus asset and check counts
New Findings — Last 7 Days	A trend line showing how many new findings appeared each day
Severity Breakdown	A bar chart showing the split between Critical, High, Medium, Low findings
By Provider	Finding counts per cloud provider
Top Families	Which check families are producing the most findings

12. Troubleshooting

12.1 The Browser Shows a Blank Page

This is caused by old build files from a previous version. Run:

```
./scripts/update.sh
```

Then restart the application.

12.2 Scan Fails Immediately

Check logs/scans.log for the error. The most common causes are:

Error Message	Cause	Fix
Credentials file not found	The path in Credential Ref does not exist	Check the path and correct it in Edit Connection
Is a directory, not a file	The Credential Ref points to a folder	Point to the file inside the folder (e.g. ~/.aws/credentials)
JSONDecodeError / json parse error	The file is not valid JSON (or is an INI file without the right format)	Check the file format. AWS credentials files are INI format and work directly.
Missing aws_access_key_id	The credentials file doesn't have the required field	Check the file contains aws_access_key_id and aws_secret_access_key
Access Denied / 403	The credentials don't have permission to read the resource	Check IAM permissions. The user needs at minimum ReadOnlyAccess.
Connection timed out	Network issue or wrong region	Check network connectivity and that the specified regions are correct

12.3 Scan Is Running but No Log Output

Check that the scan actually started by looking at the bottom of logs/scans.log:

```
tail -50 logs/scans.log
```

If the scan is stuck with no progress after several minutes, it may have hit a network timeout. CloudGuard Pro will automatically fail the scan after 30 minutes.

12.4 Check Catalog Shows 500 Error

This usually means there is a schema mismatch in the database. Stop the application, delete the database, and restart to rebuild it from scratch:

```
./scripts/stop.sh native

rm cspm.db

./scripts/start.sh native
```

12.5 pydantic-core Build Error During Deploy

If you see an error mentioning pyo3 or Python 3.14, the fix is to remove the virtual environment and re-run the deploy script:

```
rm -rf venv

./scripts/deploy.sh native
```

13. Quick Reference

13.1 Common Commands

```
# Deploy (first time or after updating)
./scripts/deploy.sh native

# Deploy without demo data
./scripts/deploy.sh native --no-demo

# Start (background)
./scripts/start.sh native

# Start (foreground, see logs in terminal)
./scripts/start.sh native --foreground

# Stop
./scripts/stop.sh native

# Clear old build artifacts (after updating)
./scripts/update.sh

# Follow scan log
tail -f logs/scans.log

# Follow main log
tail -f logs/cloudguard.log

# Run tests
PYTHONPATH=. pytest tests/ -v
```

13.2 File Layout

```
cspm/
├── backend/                    ← Python backend
│   ├── api/routes/             ← API endpoints
│   ├── check_engine/           ← Security check engine
│   ├── checkpacks/families/    ← Built-in check files (one per family)
│   ├── collectors/             ← Cloud API collectors
│   ├── custom_checks/store/    ← Your custom checks (YAML files)
│   └── workers/                ← Scan execution engine
├── frontend/dist/index.html    ← Web interface (self-contained)
├── logs/
│   ├── cloudguard.log          ← Main application log (rotates at 100 MB)
│   └── scans.log               ← Detailed scan log (rotates at 50 MB)
├── scripts/
│   ├── deploy.sh               ← Install dependencies
│   ├── start.sh                ← Start the application
│   ├── stop.sh                 ← Stop the application
│   └── update.sh               ← Clear stale build files
├── .env                        ← Configuration (edit this file)
├── .env.example                ← Configuration template
└── cspm.db                     ← SQLite database (created on first run)
```

13.3 Severity Definitions

Severity	Meaning	Example
Critical	Immediate risk — can lead to full account compromise or data breach	Root account MFA disabled, S3 bucket fully public
High	Serious risk that should be addressed within days	SSH open to the internet, RDS instance publicly accessible
Medium	Moderate risk — address within weeks	KMS key rotation disabled, no VPC flow logs

Severity	Meaning	Example
Low	Minor risk or hardening recommendation	Missing resource tags, unused IAM credentials
Informational	Observation — no immediate action required	Multi-region trail not enabled in all regions

14. Support and Feedback

CloudGuard Pro is built and maintained by Aniza Corp.

14.1 Reporting Issues

When reporting a problem, please include:

36. The contents of logs/scans.log (if the issue is scan-related)
37. The contents of logs/cloudguard.log
38. Your Python version: run python3 --version in the terminal
39. Your operating system and version
40. The exact steps to reproduce the problem

14.2 API Documentation

CloudGuard Pro includes a built-in interactive API reference. With the application running, open your browser and go to:

```
http://localhost:8000/docs
```

This shows all available API endpoints, request formats, and allows you to test calls directly from the browser.